

ECN 1620 Prices, Markets and Money: formula sheet

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You may bring one unmarked printed copy of this sheet to the midterm and to the final exam. Nothing may be written on it.

A. Elasticity

- Percentage change = $(\text{new} - \text{old}) / \text{average of old and new} \times 100$. This is the midpoint method. Use it for every elasticity.
- Price elasticity of demand (PED) = $\% \text{ change in quantity demanded} / \% \text{ change in price}$. Report PED as a positive number.
- Elastic: PED above 1. Unit elastic: PED equal to 1. Inelastic: PED below 1.
- Cross-price elasticity = $\% \text{ change in quantity of good A} / \% \text{ change in price of good B}$. Keep the sign. Positive: substitutes. Negative: complements.
- Income elasticity = $\% \text{ change in quantity} / \% \text{ change in income}$. Keep the sign. Negative: inferior good. Between 0 and 1: necessity. Above 1: luxury.
- Price elasticity of supply = $\% \text{ change in quantity supplied} / \% \text{ change in price}$.
- Total revenue = price x quantity.

B. Prices

- CPI in year t = $\text{cost of the basket in year t} / \text{cost of the basket in the base year} \times 100$.
- Inflation rate = $(\text{CPI this year} - \text{CPI last year}) / \text{CPI last year} \times 100$.
- Real rate = nominal rate - inflation, when inflation is below 10%. At 10% or more: $(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$.
- Real value = $\text{nominal value} / \text{CPI} \times 100$.

C. Money and policy

- Quantity equation: $M \times V = P \times Y$. Growth form: money growth + velocity growth = inflation + real output growth.
- Money multiplier = $1 / \text{reserve ratio}$.
- Classroom rule: policy rate = neutral real rate + inflation + $0.5 \times (\text{inflation} - \text{target}) + 0.25 \times \text{output gap}$. Use a neutral real rate of 1% and a target of 2% unless told otherwise.
- Rule of 70: years to double = $70 / \text{annual growth rate in percent}$.

D. Rounding

- Elasticities: 2 decimal places.
- Rates and percentages: 1 decimal place.
- Money amounts: whole crowns, unless the problem is stated in cents.
- Round only at the end. Carry at least four significant figures through the working.

E. Symbols

- P price, Q quantity, TR total revenue.
- M money supply, V velocity of circulation, Y real output.
- CPI consumer price index, with the base year set to 100.

- The output gap is actual output less potential output, as a percentage of potential. A negative gap means the economy is producing below potential.

F. Reminders

- State the method in one line before you calculate. A bare number earns no method marks.
- Give units. An interest rate is in percent a year; an elasticity has no units.
- If a problem gives you a table of prices and quantities, say which two rows you are comparing.